

EV & Auto

“Rigid Standardisation Could Slow Advancements In The EV Industry”

Unlike the automotive and home appliance sectors, where 60 to 70% of the components are standardised—driving down costs—the EV industry remains in flux. Anagh Ojha from Urja Mobility explored this with EFY’s Nitisha.



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What do you think are the key challenges hindering EV adoption in India?

EV ownership comes with significant challenges compared to gasoline cars, particularly in terms of battery costs. For instance, an electric car priced at ₹1.2 million includes a battery costing ₹500,000-600,000—almost 50-60% of the total price. While batteries typically last 150,000 km (around 4.5 years), replacing them later costs ₹400,000-500,000. In contrast, petrol cars require only fuel expenses without major capital investments over 10 years.

Does the absence of standardisation in the EV industry affect costs and innovation?

The EV industry is still evolving and lacks standardisation, unlike the automotive and home appliance sectors, where 60-70% of components are standardised, reducing costs. In EVs, every manufacturer designs unique products, increasing development expenses. While standardisation is needed to streamline costs, enforcing it too early may hinder innovation. Battery technology is rapidly advancing—from

lead-acid to lithium- and emerging solutions like sodium and solid-state batteries offer greater efficiency with lower weight. Rigid standardisation could slow these advancements, shifting focus from innovation to commercial gains and limiting the best solutions for customers.

What are the biggest challenges in scaling battery swapping and charging infrastructure to support leasing services?

India is a power-surplus country, generating enough electricity to meet demand. However, the supply gap of nearly 200 terawatt-hours exists due to transmission and distribution losses. Additionally, power generation and consumption are not always aligned, leading to wastage during peak production hours. The biggest challenge for charging infrastructure is the limited distribution infrastructure, designed initially for household consumption rather than EV charging needs. Setting up a charging station requires a minimum power supply of 30kW per location, which is difficult to obtain in Tier-2 and Tier-3 cities. Distribution networks need upgrades, and integrating energy

storage solutions at the local level can optimise infrastructure without requiring heavy capital investments.

How can the distribution infrastructure be made more flexible to support EV adoption?

The distribution infrastructure must become more flexible through energy storage solutions and digitalisation to support EV adoption. Government initiatives are pushing for digitalising electric meters, transformers, LT panels, and feeders, enabling real-time monitoring and management. Urja Mobility has piloted the vehicle-to-grid (V2G) concept, which allows energy storage at the local level, ensuring stored power is available for EV charging during peak hours. Additionally, promoting solar energy generation at the household level can reduce the grid’s burden, freeing up energy for EV infrastructure. These measures will make distribution networks more efficient and capable of handling increased demand from EVs. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/Idtalkapril25>

